Laptop Diagnostic Toolkit v6.0

Team Reference Guide — Consolidated Single-Module Architecture

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1. Overview & Architecture

The Laptop Diagnostic Toolkit v6.0 is a USB-portable, single-module PowerShell diagnostic and remediation suite designed for Lenovo ThinkPad fleet management. It runs from any USB drive with no installation required.

Architecture

Component	File	Purpose
PowerShell Module	`Laptop_Diagnostic_Suite.ps1`	All 45 functions in one file (~6,000 lines)
BAT Launcher	`Laptop_Master_Diagnostic.bat`	Admin elevation, menu UI, function dispatch
Config	`Config\config.ini`	All thresholds, paths, toggles (23 sections)

Module Registry	`DiagnosticModules.csv`	45 rows mapping functions to categories

Key Design Principles

- **Single-module** — one PS1 file, no external dependencies
- **USB-portable** — all paths use `$PSScriptRoot`, logs/reports stay on USB
- **ThinkPad-optimized** — auto-detects ThinkPad via 3 WMI sources (Model, SystemFamily, ComputerSystemProduct.Version)
- **Non-destructive by default** — diagnostic functions are read-only; troubleshooters require Y/N approval per fix
- **Enterprise-ready** — Fleet Report exports CSV for aggregation across multiple laptops

2. Quick Start

Step 1: Plug in USB

Insert the USB drive containing the toolkit into the target laptop.

Step 2: Run as Administrator

Right-click `Laptop_Master_Diagnostic.bat` and select **Run as administrator**. The BAT file handles UAC elevation automatically if needed.

Step 3: Select an Option

Choose from options 0–48 in the menu. Each option runs a specific diagnostic, troubleshooter, or advanced analysis function.

Step 4: Review Results

HTML reports are saved to `Reports\` and logs to `Logs\` on the USB drive. Reports open automatically in the default browser.

> **Tip:** For a full system assessment, run options in this order: 1 (Advanced Diagnostic) → 14 (Battery Health) → 17 (Thermal Analysis) → 27 (Security Check). For auto-fix, follow up with options 31–35 (Troubleshooters).

> **USB Safety:** Do not unplug the USB while a function is running. Wait until you return to the main menu. All fixes are applied directly to the laptop — they persist after the USB is removed.

3. Menu Reference (Options 0–48)

#	Category	Option Name	Description	Impact
1	Diagnostics	Advanced Diagnostic	Full system scan: disk, memory, CPU, updates, event logs, network	Read-Only
2	Diagnostics	BSOD Analysis	Analyze crash dumps, reliability records, problem drivers	Read-Only
3	Diagnostics	Enhanced Hardware Test	CPU stress, memory allocation, disk I/O, network, battery	Read-Only
4	Diagnostics	Custom Tests	Select specific test categories	Read-Only
5	Diagnostics	Diagnostic Analyzer	Analyze previous diagnostic log files for patterns	Read-Only
6	Repair Tools	Boot Repair	BCD check, bootrec/bcdboot, SFC, DISM repair	Caution
7	Repair Tools	BIOS Repair	BIOS version check, age analysis, update recommendations	Read-Only
8	Repair Tools	Driver Repair	Scan problem/unsigned drivers, device reset	Caution
9	Repair Tools	Software Cleanup	Scan temp files, browser caches, WU cache, startup programs	Caution
10	Fleet Mgmt	Fleet Report	System inventory with CSV export	Read-Only
11	Fleet Mgmt	Verify Scripts	Check PS1 module syntax, function count, BAT launcher	Read-Only
12	Fleet Mgmt	Compatibility Checker	PowerShell version, .NET, execution policy, admin check	Read-Only
13	Fleet Mgmt	Hardware Inventory	Full CIM hardware enumeration with CSV export	Read-Only
14	Hardware Tests	Battery Health	powercfg report, wear/cycle analysis, charge rate	Read-Only
15	Hardware Tests	Network Diagnostic	Adapters, IP config, DNS resolution, latency tests	Read-Only
16	Hardware Tests	Performance Analyzer	CPU/memory/disk benchmarks, bottleneck ID	Read-Only
17	Hardware Tests	Thermal Analysis	30-second temperature monitoring, fan status	Read-Only
18	Hardware Tests	Display Calibration	GPU info, resolution, monitor details, dead pixel guidance	Read-Only

19	Hardware Tests	Audio Diagnostic	Sound devices, audio service status, test tone	Read-Only
20	Hardware Tests	Keyboard Test	Keyboard detection, hotkey driver check, input test	Read-Only
21	Hardware Tests	TrackPoint Calibration	TrackPoint/touchpad detection, driver check	Read-Only
22	Hardware Tests	Power Settings	Power plans, sleep/lid settings, Power Manager check	Read-Only
23	System Mgmt	Secure Wipe	Double-confirm temp/cache/WU cleanup	Destructive
24	System Mgmt	Deployment Prep	Pre-deployment checklist: disk, activation, updates	Read-Only
25	System Mgmt	Update Manager	Windows Update COM check, Lenovo System Update	Read-Only
26	System Mgmt	BIOS Update	BIOS version/age check, flash utility detection	Read-Only
27	System Mgmt	Security Check	Defender, firewall, BitLocker, TPM, Secure Boot, UAC	Read-Only
28	Refurbishment	Refurb Battery Analysis	Battery cycle count, wear, health score for resale	Read-Only
29	Refurbishment	Refurb Quality Check	10-point QA checklist	Read-Only
30	Refurbishment	Refurb Thermal Analysis	5-minute CPU stress + thermal monitoring	Read-Only
31	Troubleshooters	Security Hardening	Auto-fix Defender, firewall, UAC, Windows Update, temp cleanup	Caution
32	Troubleshooters	Network Troubleshooter	Auto-fix Wi-Fi, Bluetooth, VPN, mobile network	Caution
33	Troubleshooters	BSOD Troubleshooter	Identify crash-causing driver, rollback/update, SFC/DISM	Caution
34	Troubleshooters	Input Troubleshooter	Fix Filter/Sticky Keys, keyboard driver, Fn keys, TrackPoint	Caution
35	Troubleshooters	Driver Auto-Update	Update outdated drivers via Windows Update + Lenovo Thin Installer	Caution
39	Advanced Diag.	POST Error Reader	ThinkPad POST/BIOS error codes from Event Logs	Read-Only
40	Advanced Diag.	SMART Disk Analysis	Deep SMART attribute analysis: temp, wear, errors	Read-Only

41	Advanced Diag.	CMOS Battery Check	RTC coin cell health via clock drift + Event Logs	Read-Only
42	Advanced Diag.	Machine Identity Check	Serial Number, UUID, Machine Type, Asset Tag	Read-Only
43	Advanced Diag.	TPM Health Check	TPM 2.0, ROCA vulnerability, BitLocker integration	Read-Only
44	Advanced Diag.	Win11 Readiness Check	Full Windows 11 hardware compatibility assessment	Read-Only
45	Advanced Diag.	Lenovo Vantage Check	Vantage, Commercial Vantage, System Update stack	Read-Only
46	Advanced Diag.	Memory Error Log Check	WHEA events, WMD results, BugCheck cross-reference	Read-Only
47	Advanced Diag.	Display Pixel Check	LCD pixel policy, GPU driver, backlight health	Read-Only
48	Advanced Diag.	Enterprise Readiness	Master health scorecard (0-100 weighted score)	Read-Only
36	Other	Run Interactive Menu	PS1 built-in menu (alternative to BAT menu)	N/A
37	Other	Open Logs Folder	Open Logs\ directory in File Explorer	N/A
38	Other	Open Reports Folder	Open Reports\ directory in File Explorer	N/A
0	Other	Exit	Exit the toolkit	N/A

4. Diagnostics (Options 1–5)

All diagnostic functions are **read-only** — they scan and report without making changes.

Option 1: Advanced Diagnostic

Comprehensive system scan covering disk health (S.M.A.R.T.), memory usage, CPU performance, Windows Update status, event log analysis (errors/warnings from last 7 days), and network configuration. Best for initial triage.

Option 2: BSOD Analysis

Reads `C:\Windows\Minidump\*.dmp` crash dumps, correlates with reliability records, identifies faulting drivers/modules. Maps known crash-causing drivers (ndis.sys, nvlddmkm.sys, ntfs.sys, etc.) to root causes. Pairs with Option 33 (BSOD Troubleshooter) for auto-fix.

Option 3: Enhanced Hardware Test

Multi-component stress test: CPU load generation (~10 sec), memory allocation/pattern test (100 MB), sequential disk I/O (100 MB), network adapter enumeration, and battery validation. Estimated runtime: 30 minutes.

Option 4: Custom Tests

Select specific test categories to run: storage, memory, display, network, thermal, audio, input devices. Useful for targeted testing when the issue area is already known.

Option 5: Diagnostic Analyzer

Analyzes previously generated log files from the `Logs\` folder. Identifies patterns, recurring errors, and trends across multiple diagnostic runs. No admin privileges required.

5. Repair Tools (Options 6–9)

Option 6: Boot Repair

Checks Boot Configuration Data (BCD), runs `bootrec` / `bcdboot`, followed by `sfc /scannow` and `DISM /RestoreHealth`. Use when a laptop has boot issues or corrupted system files.

Option 7: BIOS Repair

Read-only BIOS version check, age analysis, and update recommendations. Checks against known Lenovo BIOS versions for ThinkPads.

Option 8: Driver Repair

Scans for devices with problem status codes, unsigned drivers, and outdated Lenovo-specific drivers. Can reset individual devices via disable/enable cycle. Pairs with Option 35 (Driver Auto-Update) for automatic updating.

Option 9: Software Cleanup

Scans temp files, browser caches, Windows Update cache, and startup programs. Reports sizes and offers cleanup. Use before Deployment Prep (Option 24).

6. Fleet Management (Options 10–13)

Option 10: Fleet Report

Generates a system inventory report and exports it as CSV for fleet-level aggregation. Captures: serial number, model, OS version, CPU, RAM, disk health, battery status, and more.

Option 11: Verify Scripts

Validates the toolkit itself: checks PS1 module syntax (0 parse errors expected), counts functions (37 expected), verifies BAT launcher dispatch table, and validates config.ini sections.

Option 12: Compatibility Checker

Checks prerequisites: PowerShell version (5.1+ required), .NET Framework, execution policy, admin privileges, CIM/WMI subsystem health.

Option 13: Hardware Inventory

Full CIM hardware enumeration: processor, memory, disk, GPU, network adapters, battery, monitor, USB devices. Exports to CSV.

7. Hardware Tests (Options 14–22)

Option 14: Battery Health

Generates `powercfg /batteryreport`, analyzes wear percentage (design vs. actual capacity), cycle count, charge rate, and health status. Pairs with Option 28 (Refurb Battery Analysis) for resale scoring.

Option 15: Network Diagnostic

Enumerates all network adapters, checks IP configuration, DNS resolution, latency tests (ping), proxy settings, and VPN adapter status. Pairs with Option 32 (Network Troubleshooter) for auto-fix.

Option 16: Performance Analyzer

CPU/memory/disk benchmarks with bottleneck identification. Reports current resource usage and scores performance relative to expected baselines.

Option 17: Thermal Analysis

30-second temperature monitoring with 5-second sample intervals. Checks fan status where available. Warning at 75°C, critical at 85°C (configurable).

Option 18: Display Calibration

GPU information, current resolution vs. native, monitor details (make/model/size), and dead pixel test guidance.

Option 19: Audio Diagnostic

Enumerates sound devices, checks Windows Audio service status, provides test tone guidance.

Option 20: Keyboard Test

Keyboard device detection, HID driver status check, Lenovo hotkey driver verification. Interactive typing test. Pairs with Option 34 (Input Troubleshooter) for auto-fix.

Option 21: TrackPoint Calibration

TrackPoint and touchpad device detection, pointing device driver check, calibration guidance. ThinkPad-recommended.

Option 22: Power Settings

Lists active power plan, sleep/hibernate/lid settings, and checks for Lenovo Power Manager or Vantage.

8. System Management (Options 23–27)

Option 23: Secure Wipe

Double-confirmation privacy cleanup: removes temp files, browser caches, Windows Update cache, recent file lists. Use before returning a laptop to the pool. Destructive — deleted files cannot be recovered.

Option 24: Deployment Prep

Pre-deployment checklist: disk space, Windows activation status, pending updates, battery health, driver status. Generates a pass/fail checklist for handoff.

Option 25: Update Manager

Checks Windows Update via COM API, detects Lenovo System Update or Vantage presence. Reports pending updates and last update date.

Option 26: BIOS Update

Checks current BIOS version and age. Detects flash utilities in `Tools\BIOS\` on the USB. Safety checks: requires AC power, minimum 20% battery.

Option 27: Security Check

Checks: Windows Defender real-time protection, signature age, firewall profiles, BitLocker status, TPM version, Secure Boot, UAC level. Pairs with Option 31 (Security Hardening) for auto-fix.

9. Refurbishment (Options 28–30)

Functions for evaluating used/refurbished laptops for resale readiness.

Option 28: Refurb Battery Analysis

Scores battery on a 100-point scale:

Component	Weight	Green	Yellow	Red
Wear Percentage	50%	≤ 20%	≤ 35%	> 35%
Cycle Count	30%	≤ 500	≤ 800	> 800
WMI Status	20%	OK	Degraded	Replace

Verdict: **PASS** (≥75), **WARN** (≥50), **FAIL** (<50). All thresholds configurable in `config.ini` `[Refurbishment]`.

Option 29: Refurb Quality Check

10-point QA checklist: Windows activation, driver status, disk health, security settings, cleanup status, battery health, display, input devices, network, and thermal. Generates pass/fail for each point with an overall pass rate (minimum 80% configurable).

Option 30: Refurb Thermal Analysis

5-minute CPU stress test with continuous thermal monitoring at 5-second intervals. Detects throttling under sustained load. Pass ≤75°C, Warn ≤85°C, Fail >85°C.

10. Troubleshooters — Auto-Fix (Options 31–35)

> **How Troubleshooters Work:** Each troubleshooter follows the closed-loop cycle:
> Detect Issue → Show Finding → Ask Y/N → Apply Fix → Verify → Log & Report
>
> The technician controls every action via Y/N prompts. No fix is applied without explicit approval.

Option 31: Security Hardening

Pairs with Option 27 (Security Check). Detects and **fixes** security issues:

#	Check	Auto-Fix
1	Defender real-time protection disabled	`Set-MpPreference -DisableRealtimeMonitoring $false`
2	Defender signatures outdated (>7 days)	`Update-MpSignature`
3	Firewall profiles disabled	`Set-NetFirewallProfile -Enabled True`
4	UAC disabled	Set registry `EnableLUA=1`
5	Windows Update stuck/service stopped	Restart WU services, rename SoftwareDistribution
6	Temp files exceed threshold	Clean `%TEMP%` directory

Option 32: Network Troubleshooter

Pairs with Option 15 (Network Diagnostic). Sub-menu with 5 choices:

Sub-option	Fixes
[1] Wi-Fi	Enable adapter, update driver, flush DNS, set fallback DNS (8.8.8.8/1.1.1.1), reset TCP/IP stack, disable Large Send Offload, disable power management
[2] Bluetooth	Enable adapter, start bthserv service, update driver, disable BT power saving
[3] VPN	Restart VPN services (vpnagent/rasman), reset WAN Miniport, remove orphaned services, flush DNS, install VPN client from USB
[4] Mobile Network	Enable WWAN adapter, restart WwanSvc, report SIM/driver status
[5] Run All	Execute all four sub-workflows sequentially

Option 33: BSOD Troubleshooter

Pairs with Option 2 (BSOD Analysis). Reads crash dumps, identifies the faulting driver, and applies targeted fixes:

Faulting Module	Root Cause	Auto-Fix
ndis.sys / nwifi.sys	Network driver	Update/rollback + disable LSO
nvlddmkm.sys	NVIDIA GPU	Update GPU driver (reboot warning)
atikmdag.sys	AMD GPU	Update GPU driver (reboot warning)
ntfs.sys / ntoskrnl.exe	System/disk	SFC + DISM + check disk health
tcpip.sys	Network stack	netsh reset
dxgkrnl.sys	Display	Update GPU driver
storport.sys	Storage	Check SSD health + update storage driver

Also offers: SFC scan, DISM repair, Windows Memory Diagnostic scheduling, and thermal check.

Option 34: Input Troubleshooter

Pairs with Options 20 (Keyboard Test) and 21 (TrackPoint Calibration). 8-step check:

#	Check	Auto-Fix
1	Filter Keys enabled	Set registry Flags=126
2	Sticky Keys active	Set registry Flags=506 (uses bitmask detection — only flags when actively ON)
3	Toggle Keys enabled	Set registry Flags=62
4	Keyboard layout	Report current layout (informational)
5	Keyboard driver error	Disable/enable cycle (only flags Error/Degraded status, groups duplicate HID devices)
6	Lenovo Hotkey driver missing	Run Lenovo Thin Installer from USB if available, otherwise report
7	Pointing device error	Disable/enable cycle (only flags Error/Degraded status, groups duplicate HID devices)
8	Interactive typing test	User types test phrase to verify keyboard input

v6.0 Improvements (Input Troubleshooter):

- HID devices with "Unknown" status are now correctly recognized as normal (composite USB devices) and skipped
- Duplicate HID devices are grouped — one Y/N prompt per device type instead of per instance
- Sticky Keys detection uses bitmask (bit 2) instead of string comparison, eliminating false positives
- Error output from device reset operations is cleanly caught and displayed instead of showing raw PowerShell errors

- Lenovo Hotkey driver fix can now launch Thin Installer directly from USB

Option 35: Driver Auto-Update

Pairs with Option 8 (Driver Repair). Multi-phase driver update:

Phase	Action
1. Scan	Enumerate all PnP devices, find: status != OK, driver >12 months old, unsigned
2. Categorize	Safe (auto-update OK), Caution (GPU — needs reboot), Blocked (BIOS/Firmware — never auto)
3. Windows Update	Search for driver updates via COM API, install with Y/N per driver
4. Lenovo Thin Installer	If ThinkPad + Thin Installer on USB → Y/N → silent install
5. Verify	Re-scan devices, compare before/after, show changes

Never Auto-Updated: BIOS and Firmware updates are always blocked from automatic update. The `NeverAutoUpdate` setting in config.ini controls this.

10b. Advanced Diagnostics (Options 39–48)

These 10 modules provide deep-dive enterprise diagnostics covering POST errors, SMART analysis, TPM health, Windows 11 readiness, and a consolidated enterprise scorecard. All are **read-only**.

Option 39: POST Error Reader

Scans both **System** and **Application** Event Logs for ThinkPad POST/BIOS error codes using a built-in Lenovo error code database (0175–2201). Queries events from providers: `Lenovo*`, `BIOS*`, `ACPI*`, `LenovoBiosWmi*`. Also checks kernel power events (unexpected shutdowns, BugChecks) and BIOS age. Runs 5 checks total.

POST Code Range	Category	Common Resolution
0175–0176	Security / CRC error	System board replacement
0187–0192	Machine identity	System board / VPD repair
0210	Stuck key	Replace keyboard
0270–0271	RTC / CMOS	Replace CMOS battery
2100–2110	HDD/SSD detection	Reseat/replace storage device
2200–2201	Machine Type/UUID	Set via BIOS Setup or VPD tool

Option 40: SMART Disk Analysis

Deep SMART attribute analysis via `Get-StorageReliabilityCounter`. For each physical disk, checks: read errors, write errors, uncorrectable errors, corrected read errors, start/stop cycle count, temperature, SSD wear level, and power-on hours. Known bad SSD check (UMIS LENSE series on Win11). Also reports free space per logical disk.

Attribute	Warning	Critical/Fail
Read Errors Total	≥ 1	≥ 10
Read Errors Corrected	≥ 100	—
Uncorrectable Errors	—	≥ 1
Temperature	≥ 55°C	≥ 65°C
SSD Wear	≥ 70%	≥ 90%
Power-On Hours	≥ 5 years	—
Start/Stop Cycles	Logged for reference	
Free Space	≤ 20%	≤ 10%

Option 41: CMOS Battery Check

Detects CMOS/RTC coin cell battery failure without relying on `Win32_Battery` (which only reads the main battery). Uses two detection methods:

- **NTP clock drift:** Queries `w32tm /stripchart` to measure time offset from `time.windows.com`. Warning at >120s, Fail at >3600s.
- **Event Log analysis:** Scans Kernel-General (ID 1/12) and W32Time (ID 129/134/142) for time synchronization failures and large time corrections in the last 180 days.

> **Note:** A dead CMOS battery is a $3 hardware fix (CR2032 coin cell replacement) that takes about 5 minutes. This module detects it so the technician knows to physically replace it.

Option 42: Machine Identity Check

Validates machine identity fields critical for fleet management: Serial Number, UUID, Machine Type, and Asset Tag. Detects placeholder values ("To Be Filled", "Default string", all-zeros UUID, all-F's UUID) that indicate failed VPD programming or system board replacement without identity restore.

Option 43: TPM Health Check

Comprehensive TPM assessment:

Check	Method
TPM Presence & Version	`Get-Tpm` cmdlet
TPM Firmware Version	`Win32_Tpm` (root/cimv2/Security/MicrosoftTpm)
ROCA Vulnerability (CVE-2017-15361)	Infineon TPM firmware < 7.85 detection
Event Log Issues	Event IDs 1794, 1796 (self-test failed), 5827, 5829
Sleep-Wake TPM Bug	Event ID 24620 (TPM failed to unlock after sleep)
BitLocker Status	`Get-BitLockerVolume` (with Home edition guard)
Secure Boot	`Confirm-SecureBootUEFI`

Option 44: Win11 Readiness Check

Full Windows 11 hardware compatibility assessment covering 9 requirements:

#	Requirement	Check
1	Current OS	Detects Win10 EOL status or existing Win11
2	TPM 2.0	`Get-Tpm` SpecVersion starts with "2."
3	CPU Generation	Intel 8th Gen+, Core Ultra, AMD Ryzen 2000+, Snapdragon
4	RAM	≥ 4 GB
5	Storage	System drive ≥ 64 GB
6	Secure Boot	`Confirm-SecureBootUEFI`
7	UEFI Firmware	`$env:firmware_type` with registry fallback
8	Display	≥ 720p (1280x720)

9	Internet Connectivity	`Test-NetConnection` to update.microsoft.com:443

Verdict: **READY** (all pass), **NOT READY** (failures present), or **ALREADY ON WIN11**. Also reports Windows 10 end-of-life status (Oct 14, 2025 has already passed). Upgrade priority score 1–5.

Option 45: Lenovo Vantage Check

Detects the full Lenovo software stack for enterprise environments (9 checks):

- **Commercial Vantage:** AppX package `E046963F.LenovoSettingsforEnterprise`
- **Consumer Vantage:** AppX package `E046963F.LenovoCompanion`
- **System Update:** Registry `HKLM:\SOFTWARE\WOW6432Node\Lenovo\System Update`
- **Thin Installer:** USB-based `ThinInstaller.exe` in `Tools\LenovoThinInstaller\`
- **ImController Service:** Lenovo background service for driver management
- **Lenovo Diagnostics:** AppX or desktop install at `C:\Program Files\Lenovo\Lenovo Diagnostics\`
- **Lenovo Service Bridge:** Web-based support integration service
- **Warranty Info:** Registry `HKLM:\SOFTWARE\Lenovo\Commercial Vantage\Warranty` — status and end date

Flags conflicts (e.g., both consumer and commercial Vantage installed). Recommends Commercial Vantage for enterprise ThinkPads.

Option 46: Memory Error Log Check

Cross-references multiple memory error sources:

Source	Events
Windows Memory Diagnostic	`MemoryDiagnostics-Results` Event ID 1001/1002
WHEA Logger	Event ID 17 (correctable), 18/19 (uncorrectable)
BugCheck Cross-Reference	0x1A, 0x4E, 0x50, 0xD1, 0x7A, 0xF4
Physical Memory Config	Mismatched speeds, single-channel detection
Current Usage	>90% = Warning

Option 47: Display Pixel Check

Display and GPU health assessment (7 checks):

- **GPU info:** `Win32_VideoController` — driver date, error codes, adapter RAM
- **Display PnP status:** Device health via `Get-PnpDevice -Class Display`
- **GPU TDR events:** System log Event IDs 4101/4116
- **Monitor info:** WMI monitor parameters (brightness, size, type)
- **Native resolution match:** Compares current resolution to `VideoModeDescription` native resolution — mismatch indicates scaling or driver issue
- **Interactive pixel test:** Prompts technician to enter dead pixel count (`Read-Host`). Evaluates against Lenovo 3-pixel defect threshold (ISO 9241-307 Class II)

- **Backlight bleed test:** Prompts technician to rate backlight bleed (None/Minor/Moderate/Severe). Maps to Pass/Info/Warning/Fail. Both prompts gracefully skip in non-interactive mode.

Option 48: Enterprise Readiness Report

Master health scorecard that aggregates data across 9 weighted categories into a single 0–100 score:

Category	Weight	What It Checks
Security	20%	Defender, Firewall, UAC, BitLocker, TPM, Secure Boot
Hardware	15%	CPU, RAM, physical disk health
Storage	15%	Disk health status, free space, SMART errors
Battery	10%	Wear percentage, cycle count, WMI status
Drivers	10%	Problem devices, outdated drivers
OS	10%	Windows version, activation, pending updates
Identity	10%	Serial, UUID, Machine Type validity
TPM	5%	TPM presence, version, readiness
Display	5%	GPU status, TDR events

Lifecycle recommendation based on score:

Score	Rating	Recommendation
≥ 90	EXCELLENT	Deploy as-is
70–89	GOOD	Minor maintenance recommended
50–69	FAIR	Significant maintenance needed
30–49	POOR	Major repair or replacement consideration
< 30	CRITICAL	Replace or decommission

Generates a custom HTML scorecard with CSS progress bars. Additional outputs:

- **Action Plan:** Ordered list of recommended next steps generated from all Fail/Warning items, with specific toolkit options to run for each issue.
- **Enriched Fleet CSV:** Exports a row to `Enterprise_Fleet_Enriched.csv` with fields: ComputerName, SerialNumber, Model, OverallScore, SecurityScore, HardwareScore, StorageScore, BatteryScore, DriverScore, OSScore, IdentityScore, TPMScore, DisplayScore, LifecycleRecommendation, ActionItemCount, TopActionItem, ReportDate. Appends to existing file for fleet aggregation.

11. Impact Levels & Safety

Level	Meaning	Options
Read-Only	No system changes. Safe to run anytime.	1–5, 7, 10–22, 24–30, 39–48
Caution	Makes system changes with Y/N prompts. Reversible in most cases.	6, 8, 9, 31–35
Destructive	Deletes files permanently. Double-confirmation required.	23

Safety Features

- **Y/N per fix** — troubleshooters ask before every action
- **Invoke-FixAndVerify** — every fix is verified after application; result is Pass/Partial/Fail
- **Error suppression** — non-terminating PowerShell errors are caught cleanly via `$ErrorActionPreference = 'Stop'` in the FixAndVerify wrapper, preventing raw red error text from alarming technicians
- **ThinkPad detection** — uses 3 WMI sources (Model, SystemFamily, ComputerSystemProduct.Version) for reliable detection
- **Admin elevation** — BAT launcher checks for admin and auto-elevates via UAC
- **Logging** — every action, approval, and result is logged to `Logs\` with timestamps

12. Troubleshooter Workflows

Workflow 1: Laptop Won't Connect to Wi-Fi

1. Run **Option 15** (Network Diagnostic) to identify the issue
2. Run **Option 32** (Network Troubleshooter) → select **[1] Wi-Fi Issues**
3. Approve fixes as prompted (enable adapter, flush DNS, reset TCP/IP, etc.)
4. Review the HTML report for summary

Workflow 2: Laptop Has Frequent BSODs

1. Run **Option 2** (BSOD Analysis) to identify the faulting driver
2. Run **Option 33** (BSOD Troubleshooter) to auto-fix the root cause
3. If driver update is needed, follow with **Option 35** (Driver Auto-Update)
4. Monitor for 48 hours

Workflow 3: Keyboard Not Working Properly

1. Run **Option 20** (Keyboard Test) to diagnose
2. Run **Option 34** (Input Troubleshooter) to auto-fix
3. The troubleshooter will check accessibility settings, drivers, and Fn keys
4. If hardware damage is suspected, the typing test (step 8) will help confirm

Workflow 4: Preparing a Laptop for Resale

1. Run **Option 28** (Refurb Battery Analysis) — check battery score
2. Run **Option 30** (Refurb Thermal Analysis) — check thermals under load
3. Run **Option 29** (Refurb Quality Check) — 10-point QA
4. Fix any issues: **Option 31** (Security) → **Option 35** (Drivers)
5. Final cleanup: **Option 23** (Secure Wipe) → **Option 24** (Deployment Prep)

Workflow 5: Full System Hardening

1. Run **Option 27** (Security Check) to identify security gaps
2. Run **Option 31** (Security Hardening) to auto-fix all gaps
3. Run **Option 35** (Driver Auto-Update) to patch outdated drivers
4. Review reports for anything flagged as manual-only

Workflow 6: Enterprise Fleet Assessment

1. Run **Option 48** (Enterprise Readiness Report) for the overall health score
2. For machines scoring <70, investigate low categories:
3. — Security issues: **Option 43** (TPM) → **Option 31** (Security Hardening)
4. — Storage issues: **Option 40** (SMART Disk Analysis)
5. — Memory issues: **Option 46** (Memory Error Log Check)
6. — Identity issues: **Option 42** (Machine Identity Check)
7. Check Win11 upgrade path: **Option 44** (Win11 Readiness Check)

8. Ensure Lenovo software stack: **Option 45** (Lenovo Vantage Check)

Workflow 7: Post-Error Code Investigation

1. Run **Option 39** (POST Error Reader) to identify any logged POST/BIOS errors
2. If CMOS/RTC errors (0270/0271): run **Option 41** (CMOS Battery Check) to confirm
3. If HDD errors (2100/2110): run **Option 40** (SMART Disk Analysis) for detailed disk health
4. If identity errors (2200/2201): run **Option 42** (Machine Identity Check)

13. Configuration Reference

All settings are in `Config\config.ini`. Key sections:

[General]

Key	Default	Description
Version	6.0.0	Toolkit version
ScriptFile	Laptop_Diagnostic_Suite.ps1	Main PS1 module filename

[HardwareTests]

Key	Default	Description
ThermalWarningThresholdC	75	Warning temperature (°C)
ThermalCriticalThresholdC	85	Critical temperature (°C)
CPUStressTestDurationSec	10	CPU stress test duration
ThermalMonitorDurationSec	30	Thermal monitoring window

[Refurbishment]

Key	Default	Description
CycleCountGreen	500	Battery cycles: green threshold
CycleCountYellow	800	Battery cycles: yellow threshold
WearPercentGreen	20	Battery wear: green threshold (%)
WearPercentYellow	35	Battery wear: yellow threshold (%)
MinimumBatteryScore	75	Minimum score for PASS verdict
StressDurationSec	300	Thermal stress test length (sec)

[AutoRemediation]

Key	Default	Description
AutoRemediationEnabled	1	Master switch for troubleshooters
CreateRestorePointBeforeFix	true	Create System Restore point before fixes
FallbackDNSPrimary	8.8.8.8	Fallback DNS server 1
FallbackDNSSecondary	1.1.1.1	Fallback DNS server 2
TempCleanupThresholdMB	500	Temp cleanup trigger size (MB)
WindowsUpdateMaxAgeDays	30	Days before WU is flagged as stale

DefenderSignatureMaxAgeDays	7	Days before Defender sigs flagged
LenovoThinInstallerPath	Tools\LenovoThinInstaller	Path to Thin Installer on USB
VPNInstallerPath	Tools\VPNInstaller	Path to VPN installer on USB
DriverUpdateSource	Both	WindowsUpdate, LenovoThinInstaller, or Both
NeverAutoUpdate	BIOS,Firmware	Categories blocked from auto-update

[SMARTDiskAnalysis]

Key	Default	Description
SMARTTemperatureWarnC	55	Disk temperature warning (°C)
SMARTTemperatureCritC	65	Disk temperature critical (°C)
SMARTWearWarnPercent	70	SSD wear warning (%)
SMARTWearFailPercent	90	SSD wear fail (%)
DiskFreeSpaceWarnPercent	20	Free space warning (%)

[EnterpriseReadinessReport]

Key	Default	Description
WeightSecurity	20	Security category weight
WeightHardware	15	Hardware category weight
WeightStorage	15	Storage category weight
WeightBattery	10	Battery category weight
WeightDrivers	10	Drivers category weight
WeightOS	10	OS category weight
WeightIdentity	10	Identity category weight
WeightTPM	5	TPM category weight
WeightDisplay	5	Display category weight
PassingScoreThreshold	70	Score for GOOD rating
WarningScoreThreshold	50	Score for FAIR rating

14. USB Directory Structure

```
E:\ (USB Drive)
+-- LDT-v6.0\
    +-- Laptop_Diagnostic_Suite.ps1    (Main PS1 module - 45 functions)
    +-- Laptop_Master_Diagnostic.bat   (BAT launcher with menu)
    +-- DiagnosticModules.csv          (Function registry)
    +-- Config\
    |   +-- config.ini                 (All configuration settings)
    +-- Logs\                          (Auto-created log files)
    +-- Reports\                       (Auto-created HTML reports)
    +-- Results\                       (Fleet CSV exports)
    +-- Temp\                          (Working files, auto-cleaned)
    +-- Backups\                       (Pre-repair backups)
    +-- Data\                          (Reference data)
    +-- Tools\
    |   +-- LenovoThinInstaller\       (Place ThinInstaller.exe here)
    |   |   +-- README.txt
    |   +-- VPNInstaller\              (Place AnyConnect installer here)
    |       +-- README.txt
    +-- Docs\
        +-- Laptop_Toolkit_v6_Team_Guide.pdf
```

Tools Setup

> **Lenovo Thin Installer:** Download from `support.lenovo.com/solutions/ht037099` . Place the EXE in `Tools\LenovoThinInstaller\` . Used by Options 34 (Input Troubleshooter — hotkey driver) and 35 (Driver Auto-Update).

> **Cisco AnyConnect:** Place the MSI/EXE installer in `Tools\VPNInstaller\` . Used by Option 32 (Network Troubleshooter — VPN sub-workflow) for silent VPN client installation/repair.

15. Quick Lookup Table

Find the right option for the symptom:

Symptom	Diagnose	Auto-Fix
Laptop crashes / BSOD	Option 2 (BSOD Analysis)	Option 33 (BSOD Troubleshooter)
Wi-Fi not connecting	Option 15 (Network Diagnostic)	Option 32 (Network Troubleshooter)
Bluetooth not working	Option 15 (Network Diagnostic)	Option 32 (Network Troubleshooter)
VPN broken / missing	Option 15 (Network Diagnostic)	Option 32 (Network Troubleshooter)
Keyboard typing wrong chars	Option 20 (Keyboard Test)	Option 34 (Input Troubleshooter)
Fn keys not working	Option 20 (Keyboard Test)	Option 34 (Input Troubleshooter)
TrackPoint/touchpad issues	Option 21 (TrackPoint Cal.)	Option 34 (Input Troubleshooter)
Outdated drivers	Option 8 (Driver Repair)	Option 35 (Driver Auto-Update)
Security gaps / Defender off	Option 27 (Security Check)	Option 31 (Security Hardening)
Laptop overheating	Option 17 (Thermal Analysis)	Check fans / thermal paste (manual)
Battery degraded	Option 14 (Battery Health)	Hardware replacement (manual)
Ready for resale?	Options 28, 29, 30 (Refurb suite)	Options 31, 35 then 23, 24
Boot failure	Option 1 (Advanced Diagnostic)	Option 6 (Boot Repair)
Slow performance	Option 16 (Performance Analyzer)	Option 9 (Software Cleanup)
POST beep code / BIOS error	Option 39 (POST Error Reader)	Refer to Lenovo HMM
Disk SMART errors / SSD wear	Option 40 (SMART Disk Analysis)	Replace disk (manual)
Clock resets / date wrong	Option 41 (CMOS Battery Check)	Replace CR2032 coin cell (manual)
Missing serial / bad UUID	Option 42 (Machine Identity Check)	Reprogram via VPD tool

TPM issues / ROCA vulnerability	Option 43 (TPM Health Check)	TPM firmware update / replace
Windows 11 upgrade eligibility	Option 44 (Win11 Readiness Check)	Address failing requirements
Lenovo Vantage missing	Option 45 (Lenovo Vantage Check)	Install Commercial Vantage
RAM errors / memory BSODs	Option 46 (Memory Error Log Check)	Reseat/replace DIMM (manual)
Display GPU issues	Option 47 (Display Pixel Check)	Update GPU driver / replace LCD
Enterprise health assessment	Option 48 (Enterprise Readiness)	Address low-scoring categories

16. Changelog

v6.0.0 (February 2026)

- **Troubleshooter Functions Added:** 5 new auto-fix functions (Options 31–35) with Y/N prompt per fix and verify-after-fix cycle
- **Security Hardening (31):** Auto-fix Defender, firewall, UAC, Windows Update, temp cleanup
- **Network Troubleshooter (32):** Wi-Fi, Bluetooth, VPN, mobile network with sub-menu
- **BSOD Troubleshooter (33):** Crash dump analysis → driver fix → system repair
- **Input Troubleshooter (34):** Accessibility keys, keyboard/pointing device drivers, Fn keys, Lenovo Thin Installer integration
- **Driver Auto-Update (35):** Windows Update COM API + Lenovo Thin Installer, with Safe/Caution/Blocked categorization
- **Helper Functions:** `Request-UserApproval` (consistent Y/N prompt), `Invoke-FixAndVerify` (fix + verify + clean error handling)
- **Tools Directory:** `Tools\LenovoThinInstaller\` and `Tools\VPNInstaller\` for bundled installers
- **[AutoRemediation] Config:** New config section with DNS, thresholds, driver update source, blocked categories

v6.0.0 Advanced Diagnostics (10 New Modules)

- **POST Error Reader (39):** ThinkPad POST/BIOS error code detection from Event Logs with Lenovo error code database (0175–2201)
- **SMART Disk Analysis (40):** Deep SMART attribute analysis: temperature, wear, read errors, uncorrectable sectors, known bad SSD flagging
- **CMOS Battery Check (41):** RTC coin cell failure detection via NTP clock drift and Event Log time sync errors (does not use Win32_Battery)

- **Machine Identity Check (42):** Serial Number, UUID, Machine Type, Asset Tag integrity validation for fleet management
- **TPM Health Check (43):** TPM 2.0 assessment, ROCA vulnerability (CVE-2017-15361) detection, BitLocker integration with Home edition guard
- **Win11 Readiness Check (44):** Full Windows 11 hardware compatibility: TPM 2.0, CPU generation (Intel/AMD/Snapdragon), RAM, storage, Secure Boot, UEFI
- **Lenovo Vantage Check (45):** Commercial Vantage, Consumer Vantage, System Update, Thin Installer, ImController detection and conflict checking
- **Memory Error Log Check (46):** Cross-reference WMD results, WHEA events, BugCheck codes (0x1A/0x4E/0x50/0xD1/0x7A/0xF4) for RAM failure detection
- **Display Pixel Check (47):** GPU driver status, TDR events, monitor info, Lenovo 3-pixel defect policy reference
- **Enterprise Readiness Report (48):** Master 0–100 weighted health scorecard across 9 categories with lifecycle recommendation and custom HTML scorecard
- **10 new config.ini sections** with configurable thresholds for all new modules
- **Coverage increase:** 35 → 45 functions, ~85% of all ThinkPad enterprise issues now covered

v6.0.0 Bug Fixes

- **ThinkPad Detection Fixed:** Now checks 3 WMI sources (Model, SystemFamily, ComputerSystemProduct.Version) instead of just Model. Resolves false "Non-ThinkPad" warnings on actual ThinkPads where Model returns machine type code instead of product name.
- **HID Device False Positives Fixed:** Input Troubleshooter no longer flags HID devices with "Unknown" status as problems. "Unknown" is normal for composite USB HID endpoints. Only "Error" and "Degraded" statuses trigger the fix prompt.
- **Device Grouping:** Duplicate HID devices (e.g., 3x "HID Keyboard Device") are now grouped into a single Y/N prompt per device type, instead of prompting for each instance separately.
- **Sticky Keys Bitmask:** Detection changed from string comparison (`Flags ≠ "506"`) to bitmask check (`Flags & 4 == 4`), which correctly detects only when Sticky Keys is actively ON.
- **Error Suppression:** `Invoke-FixAndVerify` now sets `$ErrorActionPreference = 'Stop'` so non-terminating CIM/PnP errors are caught cleanly instead of displaying raw red PowerShell error text.
- **Lenovo Hotkey Fix:** When the Hotkey driver is missing, the Input Troubleshooter now offers to run Lenovo Thin Installer from USB (if present) instead of just displaying a warning.